OCD-Mix: Multi-Scale Online Change-Point Detection and Quantification for High-Dimensional Temporally Dependent Streams

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***Abstract*—Online change-point detectors for high-dimensional streams are usually calibrated for independent observations. When streams are serially correlated, window sums have much larger variance than such detectors assume, and false alarms become frequent. We present OCD-Mix, an online multi-scale detector of persistent sparse or dense mean shifts in high-dimensional streams with serial dependence, together with post-alarm quantification. OCD-Mix standardises dyadic-window sums using prewhitened long-run variance estimates from a change-free training segment, with finite-window and baseline-mean corrections, and monitors two complementary charts: a maximum-norm chart for sparse shifts and a quadratic chart for dense shifts. Thresholds are calibrated by simulation, either from an archive of null streams or autonomously from each stream's own training data. After an alarm, OCD-Mix localises the change, recovers the affected coordinates, and builds a confidence interval by adapting the Chen–Wang–Samworth procedure. On a synthetic VAR(1) benchmark with diagonal dynamics (p=50), an independence-based detector raised false alarms on every null stream for $\phi \geq 0.3$, whereas OCD-Mix kept the false-alarm rate between 3.8% and 8.9% against a 5% target; autonomous self-calibration gave 7.1–9.8% on fresh streams. Detection rates were 97.6–99.4% and held-out interval coverage 96.2–98.5%. Monitoring costs $O(p|\mathcal{K}|)$ time per observation and $O(pk_{\max})$ memory. Guarantees under estimated parameters are not established, and false alarms rise to 30.5% under dense non-diagonal dynamics.**

***Index Terms*—online change-point detection, data streams, high-dimensional time series, long-run variance, temporal dependence, post-alarm inference**

I. Introduction

Continuous, high-dimensional measurement streams—cloud telemetry, sensor arrays, network counters, industrial monitoring—are the raw material of streaming analytics, and their statistical behaviour changes abruptly when hardware degrades, a link fails, a sensor is recalibrated or a workload shifts. Declaring such a change quickly, with a controlled rate of false alarms, and saying *when* it happened and *which* dimensions are affected, is a core requirement for reliable data-stream processing. Early work in the data-management community framed stream change detection as comparing windows of the stream under independence assumptions [**?**], and recent work has moved to efficient streaming segmentation of time series [**?**].

Temporal dependence breaks independence-based detectors. Online change-point detectors with strong guarantees, such as the multiscale procedure of Chen, Wang and Samworth [**?**], assume independent Gaussian observations. Real streams are autocorrelated: queues, inertia and feedback loops make consecutive measurements similar. For a coordinate with lag-one autocorrelation ϕ, the variance of a long window sum is inflated by roughly $(1+\phi)/(1-\phi)$ relative to the independent case, so a detector that standardises by the marginal variance produces inflated statistics and frequent false alarms. In our benchmark the independence-based detector raised an alarm on every null stream, after 168 steps on average at ϕ=0.3 and 8 steps at ϕ=0.8 (Sec. VII-C).

Why quantification matters. In operations, an alarm alone is rarely actionable. Operators need the change location, a confidence interval for it, and the affected dimensions. Chen et al. [**?**] showed how to provide an interval with nominal coverage and a support estimate online, but under independence.

Gap. Existing online high-dimensional detectors with post-alarm inference are developed for independent data [**?**], [**?**], [**?**], [**?**]; methods for dependent time series are either offline [**?**], low-dimensional or based on self-normalisation [**?**], or make minimal assumptions at the cost of generality [**?**]. We are not aware of a detector that combines dependence-aware standardisation, sparse and dense charts, bounded-memory monitoring and coordinate-level post-alarm inference.

OCD-Mix. We present OCD-Mix, which (i) estimates per-coordinate long-run variances by AR(1) prewhitening of a change-free training segment, (ii) converts them into a finite-window normalisation table that includes baseline-mean uncertainty, (iii) monitors dyadic windows with cumulative-sum ring buffers using a maximum chart and a quadratic chart, (iv) calibrates thresholds by simulation, optionally from the stream's own training data, and (v) after an alarm localises the change, recovers the affected coordinates and constructs an interval, adapting [**?**] to LRV-standardised, dependent data. Bounded per-observation cost and memory independent of stream length are inherited from the multiscale design of [**?**]; our contribution is making this design usable under serial dependence and quantifying where it works.

Scope. OCD-Mix targets *persistent mean shifts*. It is not a point-anomaly or subsequence-anomaly detector [**?**], [**?**], nor a concept-drift detector for model error or arbitrary distributional change [**?**], [**?**]. Its validated dependence class is *diagonal*

VAR(1) dynamics; we do not claim robustness to arbitrary α-mixing processes.

Contributions.

1) A dependence-aware standardisation for multiscale online monitoring: prewhitened Bartlett long-run variances combined with a finite-window variance table and a baseline-mean term, giving window statistics whose empirical second moment stayed in $[1.005, 1.057]$ across scales 8–512 and $\phi \in [0, 0.8]$ (Sec. V).
2) A dual-chart detector (maximum and unthresholded quadratic) with two simulation-based calibration modes, including autonomous self-calibration from the training segment, evaluated on held-out null streams with explicit reporting of the false-alarm inflation caused by parameter estimation (Secs. V-F, VII-C).
3) An adaptation of post-alarm localisation, support recovery and confidence-interval construction to LRV-standardised dependent data, with held-out empirical coverage of 96.2–98.5% against a 95% target, and a replacement of an earlier heuristic interval that undercovered (83.8% at ϕ=0) (Secs. V-H, V-J).
4) A streaming implementation analysis and a candid evaluation of limits: $O(p|\mathcal{K}|)$ monitoring time, $O(pk_{\max})$ memory, measured latency of 117–136 μs per update for $p \leq 400$, and a documented failure mode under dense non-diagonal dynamics (false-alarm rate up to 30.5%) (Secs. VI, VII).

We stress that the components are individually classical; the contribution is their integration, calibration and evaluation for dependent streams, not a new dependence-robustness theorem.

II. Related Work

A. Online and sequential change detection

Sequential change detection began with CUSUM [**?**] and its optimality theory [**?**]. For many streams, Mei [**?**] sums local CUSUM statistics, and Xie and Siegmund [**?**] use a mixture that soft-thresholds local statistics for sparse changes; both treat streams as independent. Chen, Wang and Samworth [**?**] give a multiscale procedure for high-dimensional Gaussian mean changes with patience guarantees and delay bounds that depend on sparsity, whose storage and per-observation cost are independent of the number of past observations. OCD-Mix keeps this multiscale, bounded-state architecture. Kalinke and Gavioli-Akilagun [**?**] give a non-parametric kernel detector with logarithmic time and space, optimal delay in the minimax sense, and no pre-change training data, but for independent observations. Shekhar and Ramdas [**?**] reduce detection to sequential estimation with confidence sequences, giving an average run length of at least $1/\alpha$ under minimal structural assumptions; applying it to high-dimensional dependent streams requires a valid confidence sequence for the dependent mean, which is the difficulty we address parametrically.

B. High-dimensional detection and inference

Offline theory characterises sparse high-dimensional mean-change detection: minimax rates [**?**] (with extensions to temporal dependence, primarily for dense changes), sparse alternatives [**?**], and localisation via sparse projection [**?**] or sparsified binary segmentation [**?**]. For online procedures, uncertainty quantification was lacking until Chen et al. [**?**], who produce a confidence interval with nominal coverage and length of the order of the average detection delay up to a logarithmic factor, plus a support estimate, and show that the procedure can accompany any base detector satisfying patience and delay conditions. OCD-Mix adopts this inference machinery but runs it on LRV-standardised statistics from a different base detector, so their coverage theorem does not directly apply (Sec. V-J).

C. Change detection under temporal dependence

Standardising sums of dependent data requires a long-run variance (LRV). Kernel HAC estimators [**?**], [**?**] are biased downward under strong autocorrelation; prewhitening reduces this bias but can inflate variance [**?**]. Both effects matter here: Sec. VII shows bias reduction and a residual false-alarm excess consistent with estimation variance. Dependent-data change detection includes self-normalised tests [**?**], closed-end sequential monitoring of time-series parameters that avoids LRV estimation [**?**], and offline multivariate non-parametric localisation under exponentially decaying α-mixing with a block-type LRV estimator [**?**]. Guarantees in these works do not transfer to our setting: they are offline, low-dimensional, or give no coordinate-level post-alarm inference. OCD-Mix is parametric in the dependence (per-coordinate AR(1) prewhitening) and therefore narrower in scope but supports bounded-memory, high-dimensional monitoring with attribution.

D. Streaming analytics and data engineering

The data-management community has studied change in streams from several angles. Kifer et al. [**?**] compare sliding windows with distribution-free statistics under independence. ClaSS [**?**] segments streams with self-supervised classification, with cost linear in window size and a Flink operator. Time-series anomaly detection looks for points or point sequences that deviate from a model [**?**], increasingly with learned reconstruction models such as [**?**], and concept-drift detection targets changes affecting learners [**?**], [**?**]. These tasks differ from persistent mean-shift detection: an anomaly may be transient, drift is defined relative to a model, and none provides a calibrated alarm rule with a confidence interval for the change time and an affected-coordinate set. OCD-Mix extends this line with a statistically calibrated, constant-memory detector for *mean* shifts in dependent high-dimensional streams.

III. Preliminaries and Problem Formulation

Let $X_t \in \mathbb{R}^p$, $t = 1, 2, \ldots$, be observed sequentially. We consider a single persistent mean change,

$$X_t = \mu + \theta\, \mathbf{1}_{\{t>z\}} + Z_t, \tag{1}$$

where $\mu \in \mathbb{R}^p$ is the unknown pre-change mean, $\theta \in \mathbb{R}^p \setminus \{0\}$ the change, $z \in \{m_0, m_0{+}1, \ldots\} \cup \{\infty\}$ the unknown change time ($z{=}\infty$ is a null stream), and (Z_t) a zero-mean,

TABLE I
MAIN NOTATION.

Symbol	Meaning
p, T, m_0	dimension, stream length, training length
z, θ, S, s	change time, shift, support, sparsity
$\hat{\mu}_j, \hat{\gamma}_{0,j}, \hat{r}_j$	training mean, variance, clipped AR(1) coefficient
$\hat{\omega}_j, \hat{\sigma}_j^2$	innovation LRV, prewhitened LRV
$\hat{v}_{k,j}, \hat{s}_{k,j}^2$	finite-window variance rate, window variance
$\mathcal{K} = \{k_{\min} 2^{i-1}\}$	dyadic window grid $(8, \dots, 512)$
$W_{t,k,j}$	standardised window sum
$A_{t,k}, Q_{t,k}$	maximum and quadratic charts
$\tau_A, \tau_Q,\ c_A, c_Q$	thresholds and calibrated multipliers
N, t_N	alarm time, largest violating scale
m^*, $\hat{\nu}$	localised window length, change estimate
$\hat{\jmath}$, $\hat{S}$, $\hat{S}_{\mathrm{rep}}$	anchor, CI support, reported support
$\tilde{b}_j$, t_j	signed signal scale, CUSUM tail length

strictly stationary noise process with finite second moments. The support is $S = \{j : \theta_j \neq 0\}$, $s = |S|$; $s \ll p$ is the sparse and $s \asymp p$ the dense regime. Table I lists the main symbols.

Timeline. Observations $1, \dots, m_0$ are a change-free *training* segment ($z > m_0$) from which all baseline parameters are estimated; monitoring runs for $t > m_0$ up to horizon $H = T - m_0$ ($T = 5000$, $m_0 = 1000$, $H = 4000$ in our experiments).

Dependence. Our benchmark generates $Z_t = AZ_{t-1} + \varepsilon_t$ with $\varepsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}_p(0, \Sigma_\varepsilon)$ and $\|A\|_2 < 1$, a stable Gaussian VAR(1) (hence geometrically strongly mixing; we use this only to motivate the model class, not in any derivation). The method models each coordinate by a univariate AR(1), $Z_{t,j} = r_j Z_{t-1,j} + e_{t,j}$ with $|r_j| \leq 0.95$. This is exact for the validated case $A = \phi I_p$, $\phi \in \{0, 0.3, 0.6, 0.8\}$, and an approximation otherwise; *validity is established empirically only for diagonal A.*

Objectives. (i) *False-alarm probability*: for a target α_0, $\mathbb{P}_\infty(N \leq m_0 + H) \leq \alpha_0$, with $\alpha_0 = 0.05$; this is a calibration target, not a proven bound under estimated parameters. (ii) *Detection delay*: $\mathbb{E}_z[N - z \mid N > z]$. (iii) *Localisation error*: $|\hat{\nu} - z|$. (iv) *Interval coverage*: $\mathbb{P}_z(z \in \mathrm{CI} \mid N \geq z) \geq 1 - \alpha$, $\alpha = 0.05$. (v) *Support recovery*: Jaccard index $J(\hat{S}_{\mathrm{rep}}, S) = |\hat{S}_{\mathrm{rep}} \cap S| / |\hat{S}_{\mathrm{rep}} \cup S|$. (vi) *Resources*: per-observation time and memory independent of stream length.

IV. THE OCD-MIX FRAMEWORK

Fig. 1 shows the architecture. An *offline stage* uses only the training segment: it estimates per-coordinate dependence, long-run variances and the finite-window normalisation table, and calibrates thresholds by simulating null streams (from an archive, or from a fitted model of the stream itself). An *online stage* then ingests one vector at a time, updates cumulative-sum ring buffers, evaluates both charts at all dyadic scales, and stops at the first threshold crossing. A *post-alarm stage* runs once, on the buffered recent history. The separation matters for resources: training and calibration are one-off costs, monitoring is the per-observation bottleneck, and post-alarm inference is not on the critical path. The detector has no access to ground truth: a test enforces that detection and calibration modules never read change-point labels.

V. METHODOLOGY

A. Training and baseline estimation

For each coordinate $j \in [p]$ the training segment gives

$$\hat{\mu}_j = \tfrac{1}{m_0} \textstyle\sum_{t=1}^{m_0} X_{t,j}, \quad X_{t,j}^c = X_{t,j} - \hat{\mu}_j, \tag{2}$$

$$\hat{\gamma}_{0,j} = \tfrac{1}{m_0} \textstyle\sum_{t=1}^{m_0} (X_{t,j}^c)^2, \tag{3}$$

$$\hat{r}_j = \mathrm{clip}\Big(\tfrac{\sum_{t=2}^{m_0} X_{t-1,j}^c X_{t,j}^c}{\sum_{t=2}^{m_0} (X_{t-1,j}^c)^2},\ -0.95,\ 0.95 \Big). \tag{4}$$

(2)–(3) are the sample mean and marginal variance; (4) is the least-squares lag-one coefficient. Clipping to $[-0.95, 0.95]$ prevents the variance factor $(1 - \hat{r}_j)^{-2}$ from exploding as $\hat{r}_j \to 1$. All estimates require the training segment to be change-free; contamination would bias $\hat{\mu}$ and inflate variances.

B. Prewhitened long-run variance

The long-run variance $\sigma_j^2 = \sum_{h \in \mathbb{Z}} \gamma_j(h)$ governs the variance of long window sums: $\mathrm{Var}(\sum_{i \leq k} Z_{i,j}) \approx k\sigma_j^2$. A direct Bartlett estimate truncates the autocovariance sum and is strongly biased downward when autocorrelation is high (in our tests -4.9%, -10.8%, -22.9% at $\phi = 0.3, 0.6, 0.8$ with L=20). Following Andrews and Monahan [**?**], we first filter out the dominant serial dependence. Residuals and their recentred versions are

$$\hat{e}_{t,j} = X_{t,j}^c - \hat{r}_j X_{t-1,j}^c, \qquad \tilde{e}_{t,j} = \hat{e}_{t,j} - \tfrac{1}{m_0 - 1} \textstyle\sum_{s=2}^{m_0} \hat{e}_{s,j}, \tag{5}$$

$t = 2, \dots, m_0$. These are close to white, so a short-bandwidth Bartlett estimator suffices:

$$\hat{\omega}_j = \tilde{\gamma}_{e,j}(0) + 2 \sum_{h=1}^{L_e} \Big(1 - \tfrac{h}{L_e + 1}\Big) \tilde{\gamma}_{e,j}(h), \tag{6}$$

with $\tilde{\gamma}_{e,j}(h) = \frac{1}{m_0 - 1} \sum_{t=2}^{m_0 - h} \tilde{e}_{t,j} \tilde{e}_{t+h,j}$, $L_e = 5$, floored at 10^{-8}. Recolouring through the AR(1) transfer function at frequency zero, $f_X(0) = f_e(0)/(1 - r)^2$, gives

$$\hat{\sigma}_j^2 = \hat{\omega}_j / (1 - \hat{r}_j)^2. \tag{7}$$

In tests with m_0=1000 the bias of (7) was -0.21%, -0.07% and $+0.02\%$ at $\phi = 0.3, 0.6, 0.8$ (absolute bias below 0.25%), i.e. more than an order of magnitude smaller than the direct estimate. The scalar AR(1) filter is exact for diagonal dynamics; for cross-coupled dynamics it removes only the marginal serial dependence of each coordinate (Sec. VIII). As in [**?**], reduced bias is traded against increased estimator variance, which we later observe as false-alarm inflation.

C. Finite-window variance normalisation

Using $k\hat{\sigma}_j^2$ to normalise a window sum is inaccurate for short windows, and the centring by $\hat{\mu}_j$ adds extra variance. For a stationary AR(1) with autocovariance $\gamma_0 r^h$, summing the autocovariances over a window of length k gives the identity

$$\tfrac{1}{k} \mathrm{Var}\Big(\sum_{i=1}^{k} Y_i \Big) = \sigma^2 - \frac{2\gamma_0 r(1 - r^k)}{k(1 - r)^2}, \quad \sigma^2 = \gamma_0 \tfrac{1 + r}{1 - r}. \tag{8}$$

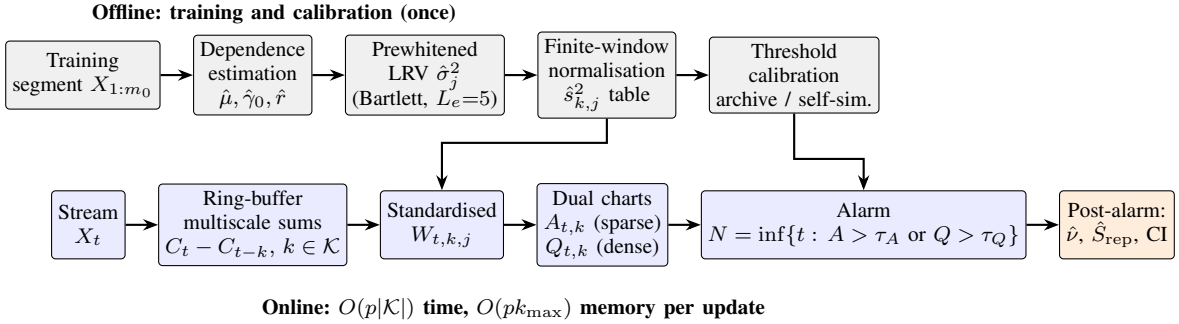


Fig. 1. OCD-Mix architecture. Grey: one-off offline stage on the change-free training segment; blue: online monitoring; orange: post-alarm inference executed once after an alarm.

OCD-Mix uses the plug-in version with the prewhitened $\hat{\sigma}_j^2$ in place of σ^2 and a floor for stability,

$$\hat{v}_{k,j} = \max\Big(\hat{\sigma}_j^2 - \frac{2\hat{\gamma}_{0,j}\hat{r}_j(1-\hat{r}_j^k)}{k(1-\hat{r}_j)^2},\ 0.2\,\hat{\gamma}_{0,j}\Big). \tag{9}$$

(8) is exact for AR(1) autocovariances with known parameters; (9) is an approximation because $\hat{\sigma}_j^2$ is estimated by a different (prewhitened) route and need not equal $\hat{\gamma}_{0,j}(1+\hat{r}_j)/(1-\hat{r}_j)$. The centred window sum $\sum_i (X_{i,j} - \hat{\mu}_j) = S_k - \frac{k}{m_0} S_{m_0}^{\text{train}}$ then has variance approximately

$$\hat{s}_{k,j}^2 = k\,\hat{v}_{k,j} + \frac{k^2\hat{\sigma}_j^2}{m_0}. \tag{10}$$

This treats the monitoring block and the training block as uncorrelated and uses $\text{Var}(S_{m_0}^{\text{train}}) \approx m_0\sigma_j^2$. Both are approximations: the cross-covariance is negligible only when the blocks are separated by more than the dependence range, and for windows adjacent to the training segment it is not exactly zero. We therefore regard (10) as a validated approximation, not an identity. The table $\hat{s}_{m,j}^2$ is precomputed for all integers $m \le 2k_{\max}$ (shape $(2k_{\max}{+}1) \times p$). Empirically, $\mathbb{E}[W_{t,k,j}^2]$ lay in $[1.005, 1.057]$ over $k \in \{8, 32, 128, 512\}$ and $\phi \in \{0, 0.3, 0.6, 0.8\}$ (20 null streams), within the acceptance range $[0.93, 1.10]$.

D. Online multiscale monitoring

The dyadic grid $\mathcal{K} = \{k_{\min}2^{i-1}\}_{i=1}^{7} = \{8, 16, \ldots, 512\}$ covers three orders of magnitude with seven scales; for any delay $\Delta \in [k_{\min}, 2k_{\max})$ some $k \in \mathcal{K}$ satisfies $k \le \Delta < 2k$, so a window of at most twice the optimal length is always available. Larger grids add sensitivity at intermediate delays but cost time linearly in $|\mathcal{K}|$. Windows lie entirely in the monitoring period, $t - k \ge m_0$.

With centred observations $x_t = X_t - \hat{\mu}$, the running sum

$$C_{t,j} = \sum_{u=m_0}^{t} x_{u,j}, \qquad \sum_{i=t-k+1}^{t} x_{i,j} = C_{t,j} - C_{t-k,j} \tag{11}$$

yields any window sum with one vector subtraction. A circular buffer of the last $k_{\max}{+}1$ values of C_t suffices. To prevent floating-point cancellation over very long streams, the buffer is re-anchored every $k_{\max}$ steps by subtracting its oldest entry from all entries (the offset is stored separately). A second circular buffer keeps the last $2k_{\max}$ centred observations for post-alarm analysis. The streaming detector reproduced the batch statistics to relative tolerance 10^{-9}.

E. Dual-chart detection statistics

The standardised window sum is

$$W_{t,k,j} = \frac{1}{\hat{s}_{k,j}} \sum_{i=t-k+1}^{t} (X_{i,j} - \hat{\mu}_j), \qquad \hat{s}_{k,j} = \sqrt{\hat{s}_{k,j}^2}, \tag{12}$$

approximately $\mathcal{N}(0,1)$ per coordinate under the null. Two charts summarise $W_{t,k} \in \mathbb{R}^p$:

$$A_{t,k} = \max_{j \in [p]} |W_{t,k,j}|, \tag{13}$$

$$Q_{t,k} = \frac{\sum_{j=1}^{p} W_{t,k,j}^2 - p}{\sqrt{2p}}. \tag{14}$$

Chart A is sensitive to sparse shifts: one strongly shifted coordinate stands out, and unaffected coordinates add no noise. Chart Q aggregates weak shifts across many coordinates and is sensitive to dense shifts. Unlike the thresholded quadratic chart of [**?**], Q has no coordinate threshold to tune. We do not assume that Q is standard normal. If every $W_{t,k,j}$ has variance one, $\mathbb{E}[\sum_j W_j^2] = p$ regardless of cross-coordinate correlation, so $\mathbb{E}[Q] = 0$; but for jointly Gaussian W with correlation matrix R, $\text{Var}(\sum_j W_j^2) = 2\|R\|_F^2$, so

$$\text{Var}(Q_{t,k}) = \|R\|_F^2/p\ \ge 1, \tag{15}$$

with equality only for $R = I_p$. Spatial dependence therefore changes the *spread and tail* of Q, not its mean, which is why τ_Q is calibrated by simulation rather than taken from a normal approximation. (If marginal variances are mis-standardised, the mean of Q also shifts; we have not separated these effects empirically.)

F. Null distribution and threshold calibration

Let $M_A = \max_{t>m_0, k\in\mathcal{K}} A_{t,k}/\sqrt{2\ln(2p|\mathcal{K}|)}$ and $M_Q = \max_{t>m_0,k\in\mathcal{K}} Q_{t,k}$ be the null maxima over horizon H. Thresholds are

$$\tau_A = c_A\sqrt{2\ln(2p|\mathcal{K}|)}, \qquad \tau_Q = c_Q, \tag{16}$$

where the Gaussian extreme-value scale gives c_A an interpretable magnitude and c_A, c_Q absorb temporal and spatial dependence.

Archive calibration. Given $n_{\mathrm{sim}} \geq 200$ null streams (fitted and monitored exactly like a real stream), set with a Bonferroni split of $\alpha_0 = 0.05$ between charts

$$c_A = \mathrm{Quantile}_{1-\alpha_0/2}(\{M_{A,i}\}), \qquad (17)$$
$$c_Q = \mathrm{Quantile}_{1-\alpha_0/2}(\{M_{Q,i}\}).$$

By the union bound this controls the probability that either chart crosses over the horizon at α_0 under the simulated law, up to Monte Carlo error. A joint variant (marginal quantiles followed by a common rescaling) differed by less than 2.3%. In our experiments the archives were generated at the dependence level of the evaluation streams.

Autonomous self-calibration. When no archive exists, the detector fits $\hat{r}$ and the innovation covariance $\hat{\Sigma}_\varepsilon = \frac{1}{m_0-1}\sum_t \tilde{e}_t\tilde{e}_t^\top$ from its own training segment, symmetrises and regularises it ($\frac{1}{2}(\hat{\Sigma}+\hat{\Sigma}^\top)+\delta I_p$), forms $\hat{\Gamma}_{0,ij} = \hat{\Sigma}_{\varepsilon,ij}/(1-\hat{r}_i\hat{r}_j)$, simulates $n_{\mathrm{sim}} = 300$ null streams of length T via $Z_0 \sim \mathcal{N}(0, \hat{\Gamma}_0)$, $Z_t = \hat{r} \odot Z_{t-1} + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, \hat{\Sigma}_\varepsilon)$, and applies (17). This is a parametric bootstrap: it inherits the estimation error of $\hat{r}$ and $\hat{\Sigma}_\varepsilon$ and uses a diagonal dependence model. We therefore report its false-alarm rate only on fresh out-of-sample null streams, separately from the in-sample rate on the calibration streams (Sec. VII-C).

G. Sequential alarm generation

Monitoring stops at

$$N = \inf\{t > m_0 : \exists k \in \mathcal{K},\ A_{t,k} > \tau_A \text{ or } Q_{t,k} > \tau_Q\}, \qquad (18)$$

so a crossing of either chart at any scale raises the alarm. At N several scales may exceed their thresholds; the largest is

$$t_N = \max\{k \in \mathcal{K} : A_{N,k} > \tau_A \text{ or } Q_{N,k} > \tau_Q\}. \qquad (19)$$

The largest violating scale is the one whose window reaches furthest back, and hence the best dyadic indication of the change's age; it bounds the post-alarm search.

H. Post-alarm changepoint localisation

The dyadic grid limits m to powers of two, so after an alarm OCD-Mix scans every integer window length $m \in [k_{\min}, \min(2t_N, N - m_0 + 1)]$ on the buffered observations (hence the $2k_{\max}$ buffer) and picks

$$m^* = \arg\max_m \max\Big(\frac{A_{N,m}}{\tau_A}, \frac{Q_{N,m}}{\tau_Q}\Big), \qquad \hat{\nu} = N - m^* + 1, \qquad (20)$$

where $A_{N,m}, Q_{N,m}$ are computed from (12) with the table entry $\hat{s}^2_{m,j}$. The criterion selects the window length at which the stronger chart is most extreme relative to its threshold; the change is estimated at its left end.

I. Coordinate support recovery

The anchor is the coordinate with the strongest response at the localised scale,

$$\hat{\jmath} = \arg\max_{j \in [p]} |W_{N,m^*,j}|. \qquad (21)$$

Because $\hat{\jmath}$ is selected by the stopping rule, its subsequent behaviour is conditioned on stopping; it is therefore treated apart from the other coordinates. With $\ell \geq 0$ extra post-alarm observations ($\ell = 0$ by default) and $k = m^* + \ell$, define $E_j = \hat{s}_{k,j}^{-1}\sum_{i=N-m^*+1}^{N+\ell}(X_{i,j} - \hat{\mu}_j)$ and $d_{1,c} = c\sqrt{\log(p/\alpha)}$. Two support sets use two constants:

$$\hat{S} = \{j \neq \hat{\jmath} :\ |E_j| - b_{\min}\sqrt{k} \geq d_{1,\mathrm{CI}}\}, \qquad (22)$$
$$\hat{S}_{\mathrm{rep}} = \{j \neq \hat{\jmath} :\ |E_j| - b_{\min}\sqrt{k} \geq d_{1,\mathrm{rep}}\} \cup \{\hat{\jmath}\}, \qquad (23)$$

$\hat{S}$ (anchor excluded, lower threshold) feeds the confidence interval; $\hat{S}_{\mathrm{rep}}$ (higher threshold, anchor always included) is what is reported to the user and scored by the Jaccard index. Here $c_{\mathrm{CI}} = 0.40$, $c_{\mathrm{rep}} = 0.80$, $b_{\min} = 0.1$, and the constants were tuned on dedicated calibration seeds, not derived from theory.

J. Post-alarm confidence interval

We adapt the interval of Chen et al. [**?**]. A dyadic grid of standardised amplitudes is $B_0 = \{b_{\min}\}$, $B = \{2^{m/2}b_{\min} : m = 1, \ldots, \lceil 2\log_2(1/b_{\min})\rceil\}$. For $j \in \hat{S}$ the signed signal-scale estimate is

$$\tilde{b}_j = \mathrm{sgn}(E_j)\max\{b \in B \cup B_0 :\ |E_j| - b\sqrt{k} \geq d_{1,\mathrm{CI}}\}. \qquad (24)$$

With $\zeta_{i,j} = (X_{i,j} - \hat{\mu}_j)/\hat{\sigma}_j$, the backward CUSUM tail length is

$$t_j = \text{smallest } \arg\max_{0 \leq h \leq h_{\max}} \Big[\tilde{b}_j \sum_{i=N-h+1}^{N} \big(\zeta_{i,j} - \tfrac{\tilde{b}_j}{2}\big)\Big], \qquad (25)$$

ties broken toward the smallest h to avoid artificial over-expansion on flat objectives, and the interval is

$$\mathrm{CI} = \Big[\max\Big(m_0,\ N - \min_{j \in \hat{S}}\big(t_j + \tfrac{4d^2_{1,\mathrm{CI}}}{\tilde{b}_j^2}\big)\Big),\ N\Big], \qquad (26)$$

and $\mathrm{CI} = [N, N]$ if $\hat{S} = \emptyset$. Each t_j estimates how long the post-change regime has lasted on coordinate j; the additive term accounts for the uncertainty of that estimate, and taking the minimum over several coordinates yields a conservative lower end. The search range $h_{\max}$ is $\min(k_{\max}, N - m_0)$ in Alg. 1; see Sec. VI for why this matters for memory.

Status of guarantees. Chen et al. prove nominal coverage for independent Gaussian data with their base detector and theoretically chosen constants [**?**]. Our setting differs in three ways: observations are dependent, standardisation uses estimated LRVs, and $c_{\mathrm{CI}}, c_{\mathrm{rep}}$ were tuned empirically. We do *not* claim a coverage theorem for the present setting; the coverage in Sec. VII is empirical, on held-out streams.

Algorithm 1 OCD-Mix: detection and post-alarm inference

Require: stream $X_t \in \mathbb{R}^p$; m_0; $\mathcal{K}$; α_0, α; $c_{\mathrm{CI}}, c_{\mathrm{rep}}, b_{\min}$

Offline stage (training segment $X_{1:m_0}$)

for $j = 1, \dots, p$ **do**
  compute $\hat{\mu}_j, \hat{\gamma}_{0,j}, \hat{r}_j$ by (2)–(4); $\hat{\sigma}_j^2$ by (5)–(7)
  for $m = 1, \dots, 2k_{\max}$: $\hat{s}_{m,j}^2 \leftarrow$ (9), (10)
end for
if self-calibration **then** simulate n_{sim}=300 null streams from fitted $(\hat{r}, \hat{\Sigma}_\varepsilon)$
else load archive of null streams
end if
$c_A, c_Q \leftarrow$ (17); $\tau_A, \tau_Q \leftarrow$ (16)

Online stage

zero buffers cum $(k_{\max}{+}1 \times p)$, obs $(2k_{\max} \times p)$
for each new X_t, $t > m_0$ **do**
  $x_t \leftarrow X_t - \hat{\mu}$; store x_t in obs; $C_t \leftarrow C_{t-1} + x_t$; store in cum
  if $t \bmod k_{\max} = 0$ **then** re-anchor cum
  end if
  $V \leftarrow \emptyset$
  for $k \in \mathcal{K}$ with $t - k \geq m_0$ **do**
    $W_{t,k} \leftarrow (C_t - C_{t-k})/\hat{s}_k$; $A_{t,k}$, $Q_{t,k}$ by (13), (14)
    if $A_{t,k} > \tau_A$ or $Q_{t,k} > \tau_Q$ **then** $V \leftarrow V \cup \{k\}$
    end if
  end for
  if $V \neq \emptyset$ **then** $N \leftarrow t$; $t_N \leftarrow \max V$; **break**
  end if
end for

Post-alarm stage (once)

m^*, $\hat{\nu}$ by (20); $\hat{\jmath}$ by (21); E_j for $k = m^* + \ell$
$\hat{S}$, $\hat{S}_{\mathrm{rep}}$ by (22), (23)
for $j \in \hat{S}$ **do** $\tilde{b}_j \leftarrow$ (24); $t_j \leftarrow$ (25) with $h_{\max} = \min(k_{\max}, N - m_0)$
end for
return $N, \hat{\nu}$, CI by (26), $\hat{S}_{\mathrm{rep}}$

VI. Algorithm and Complexity

Time. Training costs $O(m_0 p L_e)$ for estimation and $O(k_{\max} p)$ for the normalisation table. Archive calibration costs n_{sim} monitored null streams; self-calibration additionally simulates them, at $O(n_{\mathrm{sim}} T p^2)$ for correlated innovations plus $O(p^3)$ once for the Cholesky factor (our analysis; calibration runtimes were not measured). The per-observation monitoring cost is $O(p)$ to centre and update the buffers plus $O(p)$ per scale for the window difference, division, maximum and sum of squares:

$$T_{\mathrm{update}} = O(p|\mathcal{K}|), \tag{27}$$

independent of stream length T. **Memory.** The detector stores the cumulative-sum buffer ($(k_{\max}{+}1)p$ values), the observation buffer ($2k_{\max}p$) and the normalisation table ($(2k_{\max}{+}1)p$), plus $O(p)$ parameters:

$$M = O(p k_{\max}). \tag{28}$$

These bounds are for the *monitoring stage*. Post-alarm localisation is $O(p k_{\max})$ using a running sum over the observation buffer ($O(p k_{\max}^2)$ if every window sum is recomputed naively) and is executed once. The full pipeline retains the $O(p k_{\max})$ bound only if the CUSUM search range of (25) is limited to the buffered history, as in Alg. 1 ($h_{\max} \leq k_{\max} \leq 2k_{\max}$); a search over $0 \leq h \leq N - m_0$, as in a less restrictive statement of the procedure, would need the whole post-training history. We therefore state the full-pipeline memory claim conditionally on the bounded range.

TABLE II
False alarms on null streams (horizon 4000; target 0.05). OCD-Mix: archive-calibrated. ARL is censored at the horizon; ± are Monte Carlo s.e.

	OCD-Mix (archive)			Vanilla	
ϕ	FAR (N=200)	FAR (N=1000)	ARL	FAR	ARL
0.0	.020 ± .010	.038 ± .006	3961.2	.045 ± .015	3938.4
0.3	.050 ± .015	.055 ± .007	3905.7	1.000	167.7
0.6	.070 ± .018	.058 ± .007	3873.6	1.000	10.1
0.8	.100 ± .021	.089 ± .009	3799.6	1.000	8.0

VII. Experimental Evaluation

A. Setup

Streams are $X_t = \mu_t + Z_t$ with $Z_t = AZ_{t-1} + \varepsilon_t$, p=50, T=5000, m_0=1000, $\varepsilon_t \sim \mathcal{N}(0, \Sigma_\varepsilon)$ with Toeplitz $\Sigma_{\varepsilon,ij} = 0.25^{|i-j|}$, and stationary initialisation $Z_0 \sim \mathcal{N}(0, \Gamma_0)$ with $\Gamma_0 = A\Gamma_0 A^\top + \Sigma_\varepsilon$. The main benchmark uses diagonal $A = \phi I_p$, $\phi \in \{0, 0.3, 0.6, 0.8\}$. Null streams have $\mu_t = 0$. Change streams shift the mean at $z = 2500$ on $s = 5$ of 50 coordinates ($s/p = 0.10$) by $\theta_j = \pm e\sqrt{\gamma_{0,j}}$ with $e \in \{1.0, 1.5\}$, i.e. the shift is measured in marginal standard deviations. Thresholds are calibrated on dedicated seeds (200–400 streams per ϕ) and every result is evaluated on disjoint held-out seeds (1000 streams per cell, 8000 change streams in total). Nominal targets are $\alpha_0 = \alpha = 0.05$. Detection and calibration code has no access to ground truth. The reference implementation passes 27 unit tests, including batch-versus-online equivalence, byte-identical reproducibility across runs, and a blindness test. Code and scripts that regenerate every table and figure are to be supplied as anonymous supplemental material [URL to be added]. A *dense* robustness study replaces A by a random matrix scaled to $\|A\|_2 = \phi$.

B. Baselines

The only baseline with reported results is a *vanilla* online detector of the same multiscale family that standardises window sums by $\sqrt{k\hat{\gamma}_{0,j}}$, i.e. assumes independence; its false-alarm rate of 4.5% at ϕ=0 indicates thresholds appropriate for independent data. We report no numerical results for other detectors (offline segmentation, kernel, confidence-sequence or self-normalised methods), because none were supplied; Sec. VIII discusses the consequence. For intervals, the comparison is with the earlier heuristic half-width rule $a(\bar{r}) = 0.1222 + 1.1604\bar{r}$.

C. False-alarm evaluation

Table II and Fig. 2(a) give archive-calibrated results. The vanilla detector alarms on every null stream for $\phi \geq 0.3$ after 167.7, 10.1 and 8.0 steps on average. OCD-Mix stays near the 5% target: 3.8%, 5.5%, 5.8% and 8.9% at $\phi = 0, 0.3, 0.6, 0.8$ over 1000 streams, with censored average run lengths of

about 3800 or more of the 4000 monitored steps. The rate at ϕ=0.8 is several Monte Carlo standard errors above target, so control is approximate, not exact, at strong dependence. *Autonomous self-calibration* is reported separately: on 1000 fresh null streams per ϕ its false-alarm rate was between 7.1% and 9.8%. These rates, all above 5%, show the cost of calibrating from parameters estimated on 1000 observations; an earlier in-sample figure of 5.3% on the 300 calibration streams was optimistic and is not used.

D. Detection performance

Table III and Fig. 2(b) show detection rates of 97.6–99.4% in all eight cells. Delay is governed by the standardised shift and by dependence. For e=1.0 the mean delay is 14.1, 26.4, 58.7 and 144.4 steps at $\phi = 0, 0.3, 0.6, 0.8$, i.e. ratios of 1.9, 4.2 and 10.2 to the ϕ=0 value; for e=1.5 they are 6.5, 11.2, 24.0 and 56.3 (ratios 1.7, 3.7, 8.7). The scaling $(1+\phi)/(1-\phi) = 1, 1.86, 4, 9$ follows from a restricted analytic argument (a single-coordinate expected-signal-equals-threshold heuristic with the shift measured in marginal standard deviations), which gives $\mathbb{E}[N-z] \asymp \tau_A^2(1+\phi)/((1-\phi)e^2)$, and matches the measurements to within about 15%, with the largest deviation at ϕ=0.8. The ratio of delays at e=1.0 and e=1.5 (2.2, 2.4, 2.4, 2.6) is close to the predicted 1.5^2=2.25 but drifts upward with ϕ. We treat this as a supported scaling relation, not a proven expected-stopping-time theorem. The 0.6–2.4% of change streams without detection are not analysed further here.

E. Changepoint localisation

The median absolute localisation error $|\hat{\nu} - z|$ was between 0 and 1 step in every scenario with $\phi \leq 0.6$ and 4 steps at ϕ=0.8. This is small compared with the delay (up to 144 steps), showing that refining the window length after the alarm recovers most of the delay: the alarm time itself is a poor estimate of the change time, $\hat{\nu}$ a good one. Scenario-level localisation values were not tabulated in the source material, so only this range is reported.

F. Confidence intervals

Empirical coverage on held-out streams was 96.2–98.5% (Table III, Fig. 2c) against the 95% target, with the lowest values at ϕ=0.8 (96.2%) and the highest at ϕ=0.3, e=1.5 (98.5%). Mean interval width was 1.83–2.04 times the mean delay, nearly constant over ϕ and e; thus intervals scale with the difficulty of detection, as expected from the theory in the independent case [**?**], although we have not verified this scaling analytically here. By contrast, the earlier heuristic multiplier collapsed to a half-width of 2–4 steps at ϕ=0 and covered only 83.8–86.3%. Coverage above nominal indicates conservativeness; the lowest cells are the ones where dependence is strongest. *No coverage theorem is claimed for the present dependent-data setting*; c_{CI}=0.40 was tuned on calibration seeds and coverage was then measured on held-out seeds.

G. Support recovery

Table III gives the Jaccard index of the reported support $\hat{S}_{\mathrm{rep}}$ (c_{rep}=0.80): 0.58–0.80, best at ϕ=0, e=1.5 (0.801) and weakest at ϕ=0.8, e=1.0 (0.580). Using ℓ=20 additional post-alarm observations raised coordinate recall from 87.4% to 99.8% at ϕ=0 and from 74.9% to 86.0% at ϕ=0.6. In the acceptance gate, the reported-support Jaccard exceeded a tuned Bonferroni rule (z^*=2.71, the legacy procedure) by 0.08–0.13; we do not tabulate legacy Jaccards because scenario-level legacy values were not available. The CI support $\hat{S}$ (c_{CI}=0.40) is a different, more permissive set used only for intervals, and its quality was not measured.

H. Ablation and robustness

Prewhitening. Absolute LRV bias fell from 4.9–22.9% to below 0.25% (Sec. V), and the vanilla-versus-OCD-Mix contrast (Table II) confirms that dependence-aware standardisation is necessary, though it does not isolate prewhitening from the other changes. *Normalisation.* The finite-window table kept $\mathbb{E}[W^2]$ within $[1.005, 1.057]$ over all tested scales; an ablation of the table itself was not run. *Calibration.* Joint and split calibration differed by under 2.3%. *Intervals.* Replacing the heuristic multiplier by the CWS-based construction removed the under-coverage at ϕ=0.

Dense VAR. When A is a dense random matrix with $\|A\|_2 = \phi$, null false-alarm rates were $7.0 \pm 1.8\%$ (ϕ=0.3), $15.0 \pm 2.5\%$ (0.6) and $30.5 \pm 3.3\%$ (0.8) (Fig. 3). Coordinate-wise prewhitening removes the marginal serial dependence of each series but not cross-series feedback $A_{ij}Z_{t-1,j}$. A plausible mechanism is that cross-coordinate long-run correlation inflates the spread of Q via (15), so thresholds calibrated for the diagonal case are too low; we have not verified this decomposition.

I. Computational performance

Table IV and Fig. 4 report the online detector. Memory grows by 0.64 KiB between 5k and 50k updates for every p, and the footprint scales linearly in p ($\approx$ 12.1 KiB per dimension, consistent with the two ring buffers). Latency was 117–136 μs per update, i.e. over 7300 updates per second even at p=400. Latency rises by only 14% while p increases eightfold, so in this regime interpreter and per-call overhead dominate and (27) is an upper bound on arithmetic, not a measured linear trend. These numbers cover the monitoring stage only; training and calibration costs were not measured.

Implications for stream processing. The detector is one self-contained state object per monitored stream. By the measured footprint (about 12.1 KiB per dimension), a p=50 stream needs roughly 0.6 MiB, so a thousand such streams need on the order of 600 MiB of detector state, independent of how long they run; a single Python process sustained over 7000 updates per second at p=400 in our benchmark. These are arithmetic consequences of Table IV, not measurements of a streaming engine: we have not integrated OCD-Mix into one, and training and calibration would have to be scheduled outside the per-tuple path.

TABLE III
DETECTION, DELAY, INTERVAL COVERAGE AND SUPPORT RECOVERY ON 1000 HELD-OUT CHANGE STREAMS PER ROW (p=50, s=5, z=2500). CI: CWS-BASED INTERVAL; COVERAGE WITH MONTE CARLO S.E. JACCARD IS FOR $\hat{S}_{\mathrm{rep}}$ WITH c_{rep}=0.80.

ϕ	e	Detection rate (%)	Mean delay	Median delay	CI coverage (%)	Mean CI width	Width/delay	Jaccard
0.0	1.0	98.4	14.1	14.0	97.8 ± 0.46	26.9	1.91	0.770
0.0	1.5	98.6	6.5	7.0	98.0 ± 0.44	13.3	2.04	0.801
0.3	1.0	99.0	26.4	26.0	97.2 ± 0.52	49.9	1.89	0.747
0.3	1.5	98.2	11.2	11.0	98.5 ± 0.38	21.2	1.89	0.785
0.6	1.0	99.4	58.7	57.0	97.2 ± 0.52	110.9	1.89	0.711
0.6	1.5	97.9	24.0	24.0	97.9 ± 0.45	45.1	1.88	0.750
0.8	1.0	97.6	144.4	138.0	96.2 ± 0.60	263.7	1.83	0.580
0.8	1.5	97.6	56.3	55.0	96.2 ± 0.60	104.5	1.86	0.704

TABLE IV
MONITORING-STAGE COST (PYTHON/NUMPY; MEMORY IN KIB VIA TRACEMALLOC AT BUILD AND AFTER 5K AND 50K UPDATES). HARDWARE NOT RECORDED.

p	Build	5k	50k	Upd./s	μs/upd.
50	606.2	607.1	607.7	8344	119.8
100	1210.1	1211.0	1211.6	8539	117.1
200	2417.9	2418.8	2419.4	8139	122.9
400	4833.5	4834.4	4835.0	7351	136.0

J. Interpretation

Three conclusions are supported. First, dependence-aware standardisation is decisive: an independence-based detector is unusable for $\phi \geq 0.3$, while OCD-Mix stays within a few points of the target. Second, the price is real: false-alarm control is approximate (up to 8.9% archive-calibrated, 7.1–9.8% self-calibrated), delay grows by about an order of magnitude from ϕ=0 to 0.8 at fixed standardised shift, and support recovery degrades to a Jaccard of 0.58 for weak shifts at strong dependence. Third, the dependence model is a hard boundary: with dense coupling the detector over-alarms by a factor of six at ϕ=0.8.

VIII. DISCUSSION AND LIMITATIONS

Table V summarises what is and is not established. *1. Restricted dependence class.* Validation covers diagonal VAR(1). Claims of robustness to general α-mixing are not made. *2. Dense coupling.* Under non-diagonal A the false-alarm rate reached 15.0% and 30.5%; a multivariate (sparse VAR) prewhitening filter is a natural extension but is not part of the present method. *3. Estimated parameters.* Threshold calibration controls false alarms under the simulated law; theoretical finite-sample control with estimated $\hat{\sigma}_j^2$ is open, and the observed excess (to 8.9% and 9.8%) is its empirical footprint. A closed-form inflation as a function of m_0 and ϕ is not available. *4. Interval theory.* Coverage is empirical; the independent-data theorem of [**?**] does not cover dependence, estimated LRVs or tuned constants. *5. Single change.* The detector stops at the first alarm; recurrent changes require a reset protocol that re-estimates baselines after the change, which we have not built. *6. Clean training data.* Assumption of a change-free segment is essential; contamination biases $\hat{\mu}$ and the variance table. *7. Gaussian benchmark.* All evidence is for Gaussian innovations and synthetic data; heavy-tailed or real streams are untested. *8. Simulation calibration.* Thresholds rely on simulation (archive or bootstrap); archives in our experiments were matched to the dependence level. *9. Support estimator.* $\hat{S}_{\mathrm{rep}}$ lacks a consistency theorem under dependence, recall is limited without post-alarm observations (ℓ>0), and its Jaccard drops to 0.58 at ϕ=0.8, e=1.0. *10. Open mathematical issues.* The approximations in (9)–(10), the distribution of Q under spatial dependence, the full-pipeline memory bound (Sec. VI) and the delay scaling (a restricted argument) remain to be settled. The evaluation also lacks comparison with other detectors and real data. *Possible extensions*, none of which is a contribution of this paper, include sparse multivariate prewhitening, reset after alarm, and robust M-estimation for heavy tails.

TABLE V
EVIDENCE STATUS OF OCD-MIX COMPONENTS.

Component	Status
Prewhitened LRV (7)	Simulation-validated for diagonal A (bias $<$ 0.25%)
Finite-window variance (9)	Exact identity (8) for AR(1) with known parameters; plug-in is an approximation
Mean-uncertainty term (10)	Approximation (train/monitor covariance neglected); $\mathbb{E}[W^2]$ validated
Charts A, Q	Implemented; null law of the maximum of Q under dependence not derived
False-alarm control	Simulation-calibrated; no finite-sample proof with estimated parameters
Localisation (20)	Empirical (median error ≤ 1 for $\phi \leq 0.6$)
Interval (26)	Empirical coverage; theorem known only for independent data
Support (23)	Empirical; no consistency theorem under dependence
Delay scaling	Restricted analytic argument plus simulation
Complexity	Monitoring stage by inspection and benchmark; full pipeline conditional (Sec. VI)

IX. CONCLUSION

We presented OCD-Mix, an online multiscale detector of persistent mean shifts in high-dimensional streams with serial dependence, combining prewhitened long-run variance standardisation, finite-window normalisation, sparse and dense charts, simulation-based calibration and post-alarm localisa-

tion, support recovery and interval construction. On a Gaussian diagonal-VAR(1) benchmark, it avoided the complete false-alarm collapse of an independence-based detector, held false alarms to 3.8–8.9% (archive-calibrated) and 7.1–9.8% (self-calibrated) against a 5% target, detected 97.6–99.4% of changes, and gave intervals with 96.2–98.5% empirical coverage, at $O(p|\mathcal{K}|)$ time and $O(pk_{\max})$ memory per monitored observation. The evidence is empirical: false-alarm guarantees under estimated parameters and interval validity under dependence are unproven, and dense non-diagonal dynamics break the calibration. Future work should address multivariate prewhitening, theory for estimated parameters, comparison with other detectors, and real streams.

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[AUTHORS TO CONFIRM AND COMPLETE.] The text, equations typesetting, TikZ figure and plotting scripts of this manuscript were drafted by an AI system (Claude, Anthropic) from the authors' internal methodology documents; all experimental numbers were transcribed from those documents, and the AI system did not run experiments or inspect the implementation. The authors reviewed and take responsibility for the content. [Funding and other acknowledgments to be added.]

References

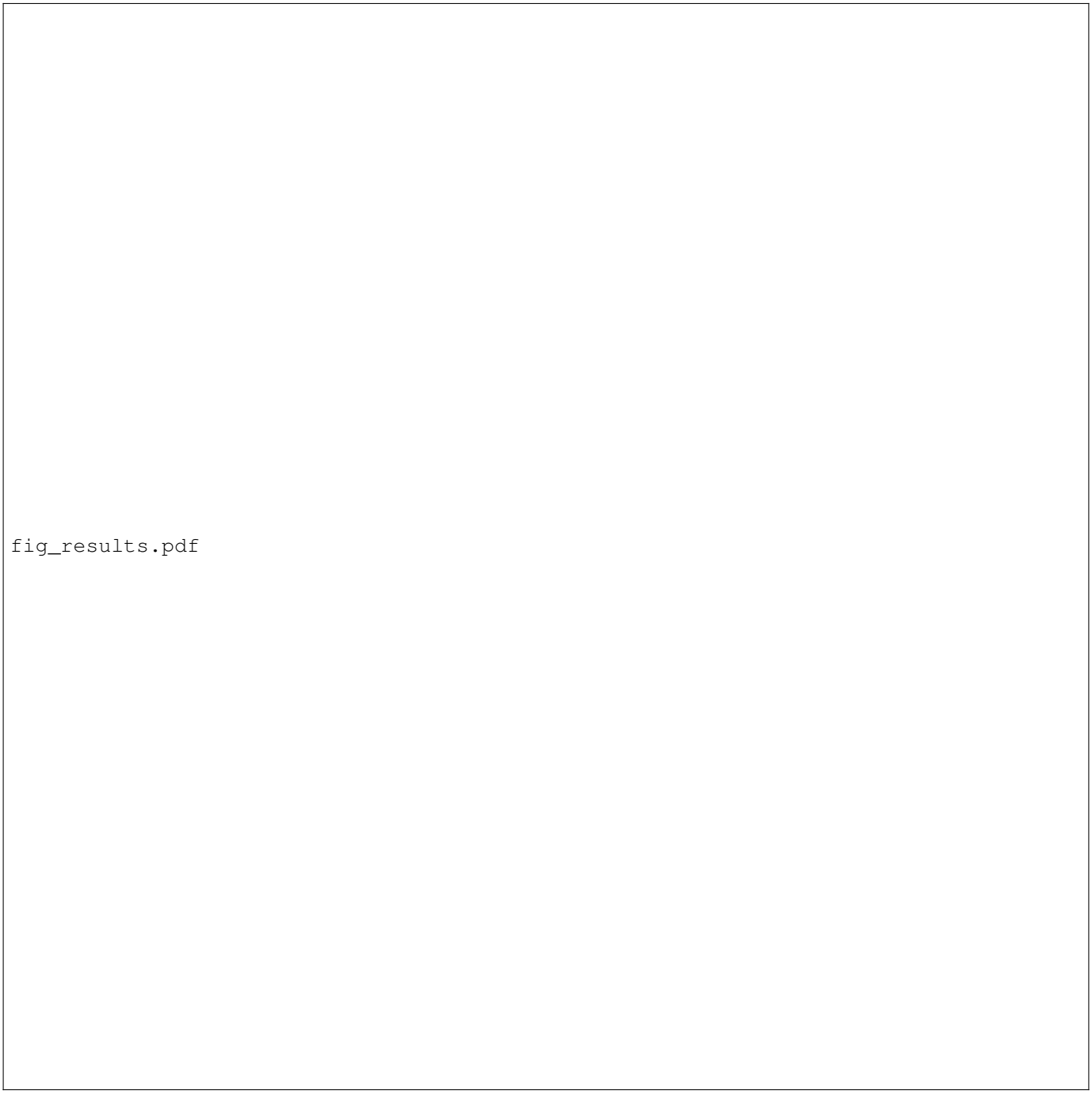


Fig. 2. (a) False-alarm rate on null streams (log scale): the independence-based vanilla detector saturates at 1 for $\phi \geq 0.3$, OCD-Mix stays close to 0.05. (b) Mean detection delay grows with ϕ roughly as $(1+\phi)/(1-\phi)$ (dashed, anchored at the measured $\phi=0$ delay; a reference, not a fit). (c) Held-out interval coverage stays above the 95% target in all eight cells. Bars: Monte Carlo standard errors.

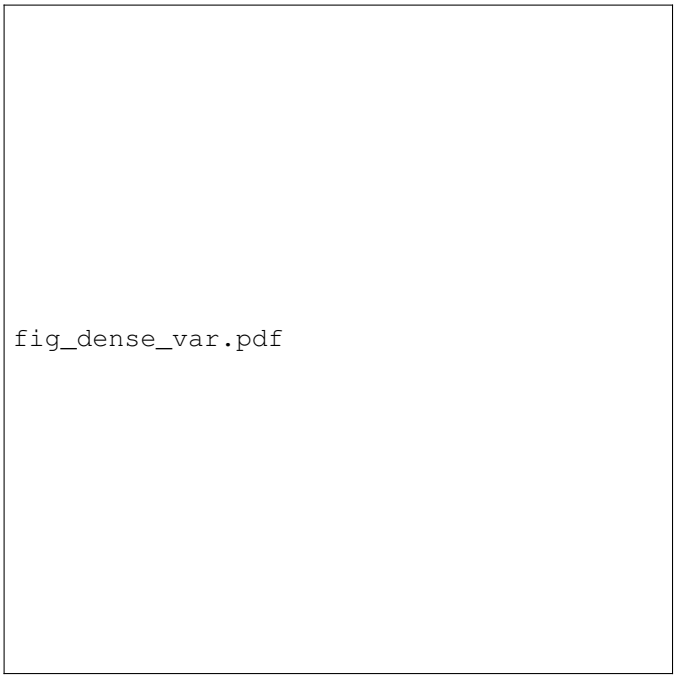


Fig. 3. False-alarm rate under diagonal versus dense transition matrices (bars: Monte Carlo standard errors). With cross-series coupling, false alarms rise to 15.0% at ϕ=0.6 and 30.5% at ϕ=0.8.



Fig. 4. Monitoring cost. (a) Live memory above the build footprint stays below 2 KiB over 50,000 updates for every p (curves overlap). (b) Latency per update grows only slowly with p.